

Project 1

I created a home lab to simulate components of a business network. The network includes the following workstations: Security Onion, Ubuntu desktop, Windows 11 Enterprise, Windows Server and lastly the attacker machine. Microsoft Active Directory, Wazuh and Mailhog are tools that will be utilized inside the environment. Future labs will be used to simulate real-world scenarios to showcase translatable skills to a real work environment. Below is a more in-depth look at each workstation and notable tools that will be utilized in future projects.

Security Onion: An open-source platform for security monitoring, log analysis, and intrusion detection, used by cybersecurity professionals to detect, investigate, and respond to network threats and incidents.

Ubuntu Desktop: General-purpose desktop. This will be used to simulate an enterprise software development environment.

Windows 11 Enterprise: Desktop operating system optimized for everyday productivity. Most common operating system used in business environments for employees. This will be used to simulate a business user.

Windows Server 2025: Designed to support enterprise-level applications and network management, and identity management. This will be used as the directory services server, acting as the central hub for network connection.

Ubuntu Server 2022: A Linux server operating system widely used for hosting applications, databases, and web services. This will be used as our email server.

Kali Linux: A Debian-based Linux distribution tailored for penetration testing and ethical hacking. It comes pre-installed with a wide range of tools for vulnerability assessment, exploitation, wireless testing, and digital forensics.

Microsoft Active Directory: A directory service used for managing and organizing network resources, users, and permissions in a Windows environment.

Wazuh: An open-source security monitoring platform that provides intrusion detection, log analysis, vulnerability detection, and compliance reporting.

MailHog: MailHog is a lightweight email testing tool that acts as a fake SMTP server. It captures all outgoing emails sent by applications, without delivering them to real inboxes.